

ENGINEERING SPECIFICATIONS

Document Name: VFD Set Points- KWI With Schneider make VFD(ATV610)

Document No:	ESP-19-20-006	Revision:	R08 (MODBUS FORMAT UPDATED FOR SIEMENS CONTROLLER)	Date: 20/10/2023
Prepared By: MRM	Revised By: IGP	Approved By: SSN		Sheet: 1 of 2

Menu	Motor Parameters	Value		
		Selection A	Selection B	Selection C
7.0 My Preferences	Access Level	Expert	Expert	Expert
4.0 Complete Setting	4.1 Motor Parameters			
	Motor Standard	60 Hz NEMA	60 Hz NEMA	60 Hz NEMA
	Nominal Motor Power KW/HP	106KW/142HP	106KW/142HP	160KW/214HP
	Nom Motor Voltage	400 V	400 V	400 V
	Nom Motor Current (A)	173 A	173 A	265 A
	Nominal Motor Frequency (Hz)	60 Hz	60 Hz	60 Hz
	Nom Motor Speed (RPM)	3509 RPM	3509 RPM	3549 RPM
	Motor Control Type	U/F VC Standard	U/F VC Standard	U/F VC Standard
	Current Limitation (A)	RLA of Comp (as per SO/Chiller nameplate) X 1.1	RLA of Comp (as per SO/Chiller nameplate) X 1.1	RLA of Comp (as per SO/Chiller nameplate) X 1.1
	Autotuning	Apply Auto Tuning	Apply Auto Tuning	Apply Auto Tuning
	Autotuning Status	Done	Done	Done
	4.2 Input/Output Menu			
	DI1 Assignment			
	DI1 High Assignment	Forward	Forward	Forward
	Ref. freq. template	Pedestal	Pedestal	Pedestal
	AI1 Config.			
	AI1 Assignment	Frequency	Frequency	Frequency
	AI1 Type	Vtg.	Vtg.	Vtg.
	AI1 Min Value	0	0	0
	AI1 Max Value	10	10	10
	R1 Config.			
	R1 Assignment	Operating State Fault	Operating State Fault	Operating State Fault
	R2 Config.			
	R2 Assignment	Drive Running	Drive Running	Drive Running
	4.3 Command & Ref.			
	Low Speed	22 Hz	22 Hz	22 Hz
	High Speed	53HZ	60HZ	60 Hz
	4.4 Generic Function			
	Ramp			
	Acceleration	8 Sec	8 Sec	8 Sec
	Deceleration	7 Sec	7 Sec	7 Sec
	Auto DC injection	No	No	No
	Threshold reached			
	High Current Thd	RLA of Comp (as per SO/Chiller nameplate) X 1.05	RLA of Comp (as per SO/Chiller nameplate) X 1.05	RLA of Comp (as per SO/Chiller nameplate) X 1.05
	Low I Threshold	0.2	0.2	0.2
	Low Freq Threshold	22	22	22
	Mot Freq Thd	53	60	60
	4.6 Error/Warning			
	Out put phase loss	YES(Can Disable only for control test)	YES(Can Disable only for control test)	YES(Can Disable only for control test)
	Warn grp configuration			
	Grp. 1 def.			
	IGBT Thermal Warn	Yes	Yes	Yes
	Fan Counter Warning	Yes	Yes	Yes
	Fan Feedback Warn	Yes	Yes	Yes
	Undervoltage Warn	Yes	Yes	Yes
	Preventive UnderV	Yes	Yes	Yes
	Mot Freq High	Yes	Yes	Yes
	Mot Freq Low Thd	Yes	Yes	Yes
	Current Thd Reached	Yes	Yes	Yes
	Drv Therm reached	Yes	Yes	Yes
	Motor Therm Thd	Yes	Yes	Yes

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1.2 Simply Start	Nominal Motor Power KW/HP	106KW/142HP	106KW/142HP	160KW/214HP
	Nom Motor Current (A)	173 A	173 A	265 A
	Motor Th Current ITH	173 A	173 A	265 A
	Acceleration	8 Sec	8 Sec	8 Sec
	Deceleration	7 Sec	7 Sec	7 Sec
	Low Speed	22 Hz	22 Hz	22 Hz
	High Speed	53 Hz	60 Hz	60 Hz
6.0 File Management	6.1 Transferconfig file	Copy to the Drive	Copy to the Drive	Copy to the Drive
		Copy from the Drive	Copy from the Drive	Copy from the Drive
7.0 My Preferences	7.2 Password			
	Password status			
	Password	202020	202020	202020
	Upload Rights	Not Allowed	Not Allowed	Not Allowed
	Download Rights	Not Allowed	Not Allowed	Not Allowed
Modbus	Modbus address (Add)	1,2 (As per circuit number)	1,2 (As per circuit number)	1,2 (As per circuit number)
	Modbus baud rate (tbr)	19K2	19K2	19K2
	Modbus format (tFo)	8 n 1 (With Carel Controller)	8 n 1 (With Carel Controller)	8 n 1 (With Carel Controller)
		8 e 1 (With Siemens Controller)	8 e 1 (With Siemens Controller)	8 e 1 (With Siemens Controller)

NOTES

**	FLA SETPOINT IN CONTROLLER
	1) FOR SELECTION 'A' - RLA(AS PER SO/ CHILER NAMEPLATE) X 0.76
	2) FOR SELECTION 'B' & 'C' - RLA(AS PER SO/ CHILER NAMEPLATE) X 0.87

SR NO	MODEL	CIRCUIT I	CIRCUIT II
1	KWI130.14	A	NA
2	KWI135.14	A	NA
3	KWI140.14	A	NA
4	KWI145.14	B	NA
5	KWI150.14	B	NA
6	KWI155.14	B	NA
7	KWI170.14/KWI130.1G	C	NA
8	KWI175.14/KWI145.1G	C	NA
9	KWI180.14/KWI170.1G	C	NA
10	KWI195.14/KWI195.1G	C	NA
11	KWI200.14/KWI210.1G	C	NA
12	KWI205.14	C	NA
13	KWI230.14	C	NA
14	KWI235.14	C	NA
15	KWI240.14	C	NA
16	KWI248.14	C	NA
17	KWI250.14	C	NA
18	KWI255.14	C	NA
19	KWI260.14	C	NA
20	KWI268.14	C	NA
21	KWI270.14	C	NA
22	KWI280.14	C	NA
23	KWI285.14	C	NA
24	KWI290.14	C	NA
25	KWI265.24	A	A
26	KWI270.24	A	A
27	KWI275.24	A	A
28	KWI298.24	B	B
29	KWI300.24	B	B
30	KWI310.24	B	B
31	KWI320.24	B	C
32	KWI325.24	B	C
33	KWI335.24	B	C
34	KWI345.24/KWI260.2G	C	C
35	KWI350.24/KWI275.2G	C	C
36	KWI355.24/KWI295.2G	C	C
37	KWI365.24/KWI320.2G	C	C
38	KWI375.24/KWI340.2G	C	C
39	KWI380.24/KWI355.2G	C	C
40	KWI395.24/KWI370.2G	C	C
41	KWI400.24/KWI380.2G	C	C
42	KWI405.24/KWI390.2G	C	C
43	KWI425.24/KWI405.2G	C	C
44	KWI430.24/KWI420.2G	C	C

45	KWI435.24	C	C
46	KWI460.24	C	C
47	KWI465.24	C	C
48	KWI470.24	C	C
49	KWI480.24	C	C
50	KWI485.24	C	C
51	KWI490.24	C	C
52	KWI500.24	C	C
53	KWI510.24	C	C
54	KWI515.24	C	C
55	KWI518.24	C	C
56	KWI520.24	C	C
57	KWI525.24	C	C
58	KWI530.24	C	C
59	KWI535.24	C	C
60	KWI540.24	C	C
61	KWI550.24	C	C
62	KWI555.24	C	C
63	KWI560.24	C	C
64	KWI565.24	C	C
65	KWI575.24	C	C
66	KWI580.24	C	C